Total No. of Printed Pages: 04



**SUBJECT CODE NO: - RNP-8018**

**FACULTY OF SCIENCE AND TECHNOLOGY**

**F.E. (Group A) (NEP) Sem - I**

**Examination November / December 2025 (To be held in February-2026)**

**Engineering Mechanics**

**[Time: 3:00 Hours]** **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.NO 2, 3, 4, 5, 6 and 7

## SECTION A

**Q1 Solve any 10 following questions** **20**

1. Force is defined as a ______ or pull acting on a body.
   A) Push B) Pressure C) Energy D) Velocity

2. The SI unit of force is ______.
   A) Dyne B) Joule C) Newton D) Watt

3. A body is said to be in equilibrium when the resultant force is ______.
   A) Maximum B) Variable C) Zero D) Infinite

4. Friction always acts in the direction ______ to motion.
   A) Same B) Parallel C) Perpendicular D) Opposite

5. Limiting friction occurs when a body is about to ______.
   A) Stop B) Rest C) Slide D) Rotate

6. Members of a truss are connected by ______ joints.
   A) Fixed B) Rigid C) Pinned D) Welded

7. Truss members carry only ______ forces.
   A) Bending B) Shear C) Axial D) Torsional

8. The path of a projectile is ______ in shape.
   A) Circular  B) Straight line  C) Elliptical  D) Parabolic

9. Acceleration is the rate of change of ______.
   A) Distance  B) Velocity  C) Displacement  D) Momentum

10. Relative velocity is the velocity of one body with respect to ______.
    A) Ground  B) Air  C) Another body  D) Itself

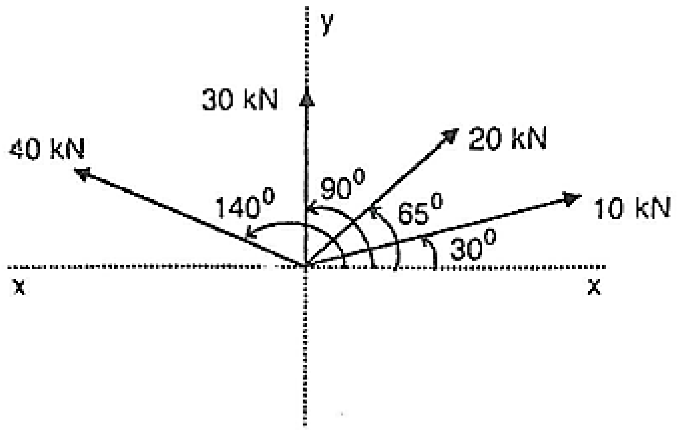
11. Momentum is equal to mass multiplied by ______.
    A) Acceleration  B) Force  C) Velocity  D) Energy

12. Impulse is equal to force multiplied by ______.
    A) Distance  B) Time  C) Velocity  D) Mass

13. Momentum remains constant when external force is ______.
    A) Applied  B) Variable  C) Zero  D) Increasing

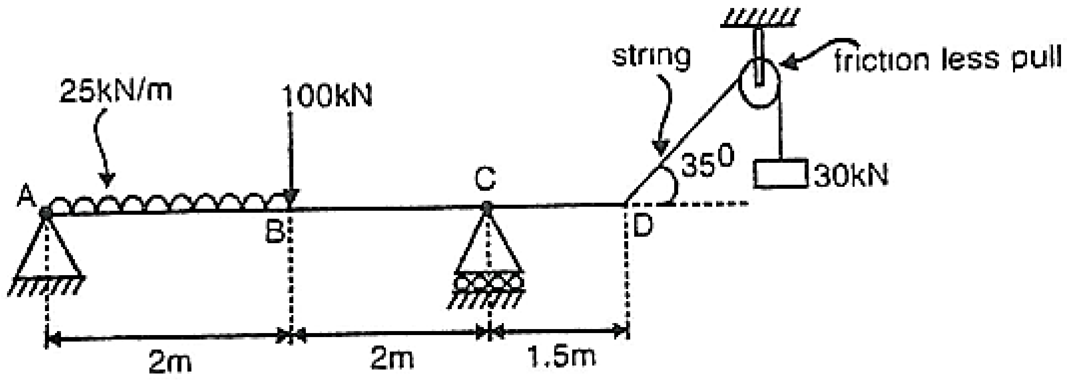
**SECTION B**

**Q2** **A)** Four forces of magnitude 10 kN, 20 kN, 30 kN and 40 kN are acting at a point as shown in fig. The angles made by these forces with x-x axis are 30°, 65°, 90° and 140° measured anticlockwise from positive x-axis. Find the magnitude and direction of the resultant force. **5**

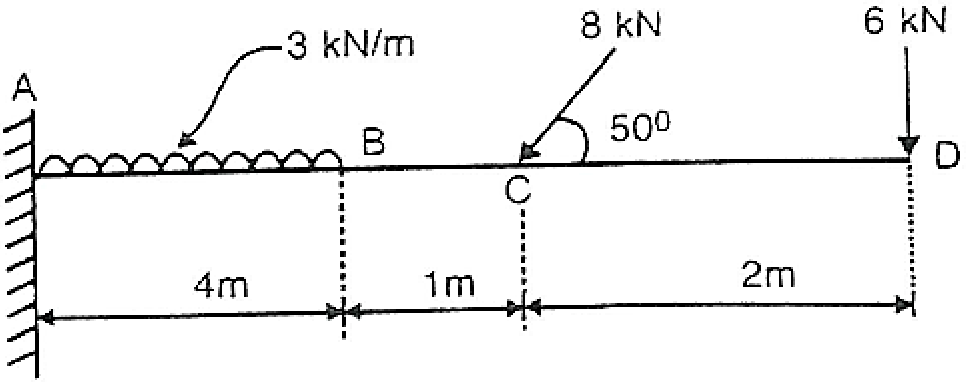



**B)** Explain the law of parallelogram theorem **5**

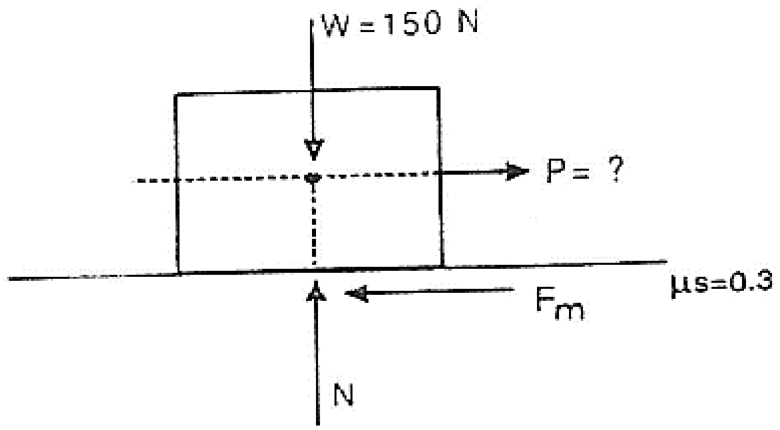
**Q3** **A)** Find the reaction at the supports for the beam loaded as shown in figure by analytical method. **5**



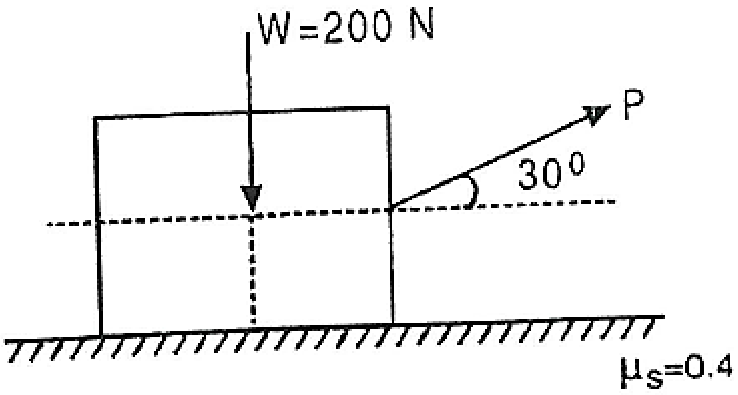

**B)** Find the reactions at the support of beam loaded as shown in figure **5**




**Q4** **A)** A body of weight 150 N is kept on rough horizontal plane. If the coefficient of friction between the body and the plane is 0.3, find the horizontal force required to just slide the body on the plane. **5**




**B)** A body of weight 200 N is kept on a rough horizontal plane and a force is applied to just move the body horizontally as shown in figure **5**
Find the magnitude of force P required if coefficient of friction is 0.4.




**Q5** **A)** On turning a corner, a motorist rushing at 20 m/s, finds a child on the road ahead. He instantly stops the engine and applies brakes, so as to stop the car within 10 m of the child. Calculate (i) retardation, and (ii) time required to stop the car. **5**

**B)** A scooter starts from rest and moves with a constant acceleration of 1.2 $m/s^2$. Determine its velocity, after it has travelled for 60 meters. **5**

**Q6** **A)** A motor car takes 10 seconds to cover 30 meters and 12 seconds to cover 42 meters. Find the uniform acceleration of the car and its velocity at the end of 15 seconds. **5**

**B)** A burglar's car had a start with an acceleration of 2 $m/s^2$. A police vigilant party came after 5 seconds and continued to chase the burglar's car with a uniform velocity of 20 m/s. Find the time taken, in which the police party will overtake the burglar's car. **5**

**Q7** **A)** A particle, starting from rest, moves in a straight line, whose equation of motion is given by: $s = t^3 - 2t^2 + 3$. Find the velocity and acceleration of the particle after 5 seconds. **5**

**B)** If a particle is projected inside a horizontal tunnel which is 5 metres high with a velocity of 60 m/s, find the angle of projection and the greatest possible range. **5**